

Sync Without Signal: Offline-First Transaction Consistency for a Multi-App POS Platform

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1. Case Context

Application K is a POS and business intelligence (BI) platform for Indonesian SMEs. It supports three operational roles through separate applications: Entry, Owner, and Operator. Among these, the Operator application is responsible for handling checkout transactions, making it the most business-critical component because it is used directly at the point of sale, where internet connectivity is often unstable.

2. Constraints

This case study follows a fixed architectural decision (**ADR-006**): the checkout application must continue operating under an offline-first onboarding model for the next six months, with no self-serve fallback. This constraint reflects the realities of Indonesian SME retail environments and should be treated as a given throughout the analysis.

Key Takeaways :

- POS + BI platform for Indonesian SMEs.
- Three applications: Entry, Owner, and Operator.
- Operator handles checkout transactions in unstable network environments.
- ADR-006 fixes offline-first onboarding for six months with no self-serve fallback.
- Teams must design their solution within this constraint.

3. Background

Application K is a POS and business intelligence (BI) platform for Indonesian SMEs, where unreliable internet connectivity is a daily operational reality rather than an exception. The checkout process must continue even during network outages to avoid disrupting sales.

The platform supports three payment methods. Cash payments are confirmed directly by the operator without relying on external systems. Static QRIS payments use a pre-generated QR code that can be displayed without an internet connection. Bank transfers are verified manually by the operator using external confirmation, such as a banking app, e-wallet notification, or SMS.

Transactions follow a simple lifecycle of Pending, Confirmed, and Voided. Once confirmed by the backend, a transaction cannot be modified by operators, while corrections are restricted to administrators as an exception workflow. Inventory is deducted only after a transaction is confirmed, reflecting an accepted level of eventual consistency rather than real-time stock reservation. In addition, multiple operators may process transactions simultaneously for the same merchant, including while working independently on different devices without network connectivity.

4. Problem Statement

Design an offline-first synchronization architecture for the Operator (POS) application that allows transactions to be created and locally confirmed while offline, then synchronized with the backend when connectivity is restored. The design must prevent duplicate or lost transactions, preserve backend-side transaction immutability, and support concurrent offline transactions from multiple devices under the same merchant.

A transaction confirmed while offline is **provisional** until accepted by the backend. This distinction between **provisional** and **settled** transactions must be explicit in both the synchronization logic and the operator UI.

Key Requirements :

- Offline transaction creation and local confirmation.
- Reliable synchronization after reconnection.
- No duplicate or lost transactions.
- Backend-side transaction immutability after sync.
- Support concurrent multi-device offline operation.
- Clearly distinguish **provisional** and **settled** transactions.

5. Data & Constraints

Business	Expected to scale from an early pilot to 500+ merchants .
	Multiple operators may process transactions simultaneously under the same merchant.
	Internet connectivity is frequently intermittent.
System Constraints	Backend-confirmed transactions are immutable.
	Cash and Static QRIS payments must remain fully operable during connectivity outages.
	Transfer payments are verified by the operator using an external confirmation

	(e.g., bank app, e-wallet, or SMS), not by the POS system.
	Inventory is deducted only after transaction confirmation. Real-time stock reservation is out of scope.
	Offline-first onboarding remains mandatory for the first six months, with no self-serve alternative.
Payment Validation	System-verifiable: Cash.
	Operator-asserted: Static QRIS and Transfer. Both rely on operator verification of an external payment signal and should be treated as carrying residual confirmation risk.

6. Deliverables

Participants should design and justify :

- An offline data persistence strategy that ensures transactions survive app restarts and device reboots.
- A synchronization protocol that reliably reconciles offline transactions while preventing duplicates.
- A concurrency strategy for multiple offline operators working under the same merchant.
- A transaction lifecycle showing how offline-created transactions transition from provisional to backend-confirmed states.
- A payment handling strategy for Cash, Static QRIS, and Transfer, including differences in their confirmation process.
- A reconciliation process for operator-confirmed payments (Static QRIS and Transfer) when payment confirmation is later found to be incorrect.
- Failure handling for concurrent synchronization, interrupted connectivity, and mass reconnection after prolonged outages.
- A justification of the chosen consistency model and its trade-offs.

7. Definition of Success

A successful solution enables Application K to support reliable checkout operations despite prolonged connectivity disruptions. Operators should be able to continue processing sales offline without losing or duplicating transactions, while all provisional transactions are reconciled correctly once connectivity returns. The system should provide clear visibility into transaction status, maintain accurate sales records across multiple operators and devices, and support Application K's growth from its initial pilot to 500+ merchants without requiring manual reconciliation as part of normal operations.

